NATALIE SIM EN XI

Data Science | Machine Learning | Data Analytics

Kuching, Sarawak, Malaysia | nataliessim@gmail.com | (+60) 111 071 3830

WORK EXPERIENCE

Maxis Broadband Sdn. Bhd. | *Mar 2026 – Aug 2026*

Data Analyst Intern, Data & AI – Applied Analytics | Menara Maxis, Kuala Lumpur

• Developed the team's first customer-level churn prediction model using Python, SQL, and XGBoost, achieving 13× lift in the first percentile and capturing 50.7% of churners in the highest-risk decile.

• Audited and optimized Google Cloud Platform (GCP) resources, including tables, notebooks, API keys and configurations, contributing to a 25% reduction in estimated monthly cloud expenditure for the Applied Analytics team.

• Conducted research and prototyping with BigQuery Graph Analytics to explore graph-based representations of complex data relationships for potential future analytics use cases.

• Enhanced a Power BI model monitoring solution integrated with production data pipelines, updating monitoring scripts, performance checks and dashboards to support post-deployment model monitoring.

EDUCATION

Universiti Malaysia Sabah (UMS) | *Oct 2022 – Nov 2026*

Bachelor of Computer Science, Data Science | Kota Kinabalu, Sabah

• CGPA: 3.85/4.00

Universitas Diponegoro | *Sep 2025 – Dec 2025*

Semester Abroad, Informatika – Diponegoro Experience Exchange Program (DEEP) | Semarang, Indonesia

St. Thomas' Secondary School | *Jun 2022*

Malaysian Higher School Certificate (STPM), Science & Mathematics | Kuching, Sarawak

• CGPA: 3.59

Kuching High School | Dec *2020*

Malaysian Certificate of Education (SPM), Science Stream | Kuching, Sarawak

• 9As

AWARDS & HONOURS

Maxis Women in Tech Scholarship

2022 Cohort | Selected among 21 scholars nationwide

Dean's List, Universiti Malaysia Sabah

All Semesters

PROJECTS & COMPETITIONS

Final Year Project: Swarm Intelligence-Optimized GNNs for Drug Repurposing

A Bio-Inspired and Adaptive Approach

• Developed and evaluated swarm intelligence approaches, including Particle Swarm Optimization (PSO) and Ant Colony Optimization (ACO), to optimize a Relational Graph Convolutional Network (RGCN) for drug repurposing on a biomedical knowledge graph.

The Sustainable Food Nexus Data Challenge | *Aug 2023*

National Level – Top 10 Finalist | Johor Bahru, Johor

• Collaborated with teammates to develop a Convolutional Neural Network (CNN) for classifying banana ripening stages.

Malaysia Techlympics 2023 – Data Science | *Dec 2023*

National Level – Top 10 Finalist | Putrajaya

• Collaborated with teammates to create a data visualization using Malaysia's Health Consumer Price Index (CPI) data to explore medical tourism in Malaysia.

ADDITIONAL EXPERIENCE

AIESEC in UMS | *Feb 2023 – Aug 2023*
Organizing Committee, Project Nature Warrior 10.0 | SDG 14 Life Below Water
• Served on the organizing committee for Youth Experience and People Development, supporting the planning and execution of the program.

Part-time English Tutor
• Provided personalized English tutoring to students from China and Taiwan during semester breaks, adapting lessons to their age and proficiency.

Freelance Website Builder
• Built websites for small businesses based on client requirements.

TECHNICAL SKILLS

Programming: Python, R, SQL
Data & Analytics: Microsoft Excel, Power BI
Cloud & Data Platforms: Google Cloud Platform, BigQuery, Vertex AI
Development & Version Control: Git, GitLab
Languages: English, Mandarin, Malay, Korean

RELEVANT COURSEWORK

Machine Learning for Data Science, Statistics and Data Analysis, Data Mining and Warehousing, Data Visualization, Database, Big Data, Artificial Intelligence, Natural Language Processing